

## Project G: Privacy-Preserving Age Verification

### Overview

In this project, you will design and implement a privacy-preserving age verification system that addresses a fundamental privacy challenge: how can you prove you're over 18 using your government-issued ID without revealing your actual age, name, address, or any other personal information? This project guides you through building a cryptographic protocol that enables selective disclosure—proving only what's necessary while keeping everything else private. You'll start by implementing a digital ID system using cryptographic commitments and signatures, then build zero-knowledge proofs that can verify age thresholds (e.g., “I am 18 or older”) without revealing the exact birthdate. The system will ensure that even repeated age checks at different venues cannot be linked together to track users. You'll tackle real-world challenges including credential revocation (what happens when an ID expires or is reported stolen?), preventing credential sharing between users, and ensuring efficient verification that works on mobile devices. The project explores the cryptographic primitives behind anonymous credentials, including Pedersen commitments for hiding attributes, Schnorr-style proofs for selective disclosure, and unlinkable signatures that prevent tracking across sessions. By completing this project, you'll understand how modern privacy-preserving identity systems work and gain hands-on experience implementing protocols that could enable age verification for online services, venue entry, or regulated purchases—all while preserving user privacy in ways that physical ID checks cannot achieve.

### Learning Objectives

After completing this project, you should be able to:

- Implement cryptographic commitments to hide personal attributes while allowing selective disclosure.
- Design and implement zero-knowledge proofs for threshold predicates (e.g.,  $\text{age} \geq 18$ ).
- Build anonymous credential systems that prevent tracking across verification sessions.
- Analyze and mitigate practical challenges including credential revocation and sharing prevention.

### Background

Privacy-preserving age verification requires careful application of cryptographic techniques to balance verification needs with privacy protection:

- Pedersen commitments allow hiding attribute values while enabling efficient zero-knowledge proofs about committed values.
- Schnorr-style proofs enable proving statements about hidden values without revealing the values themselves.
- Anonymous credentials provide unlinkability between different verification sessions, preventing tracking.
- Revocation mechanisms must balance privacy with the ability to invalidate compromised or expired credentials.
- Anti-sharing techniques prevent users from lending their credentials to others who don't meet the requirements.

## Requirements

Your privacy-preserving age verification system must implement the following core functionality:

### 1. Digital ID Issuance:

- Implement a credential issuance protocol where an authority issues digital IDs.
- Use cryptographic commitments to bind attributes (birthdate, expiry, etc.) to the credential.
- Ensure the issuer's signature validates the credential without revealing committed attributes.

### 2. Zero-Knowledge Age Proofs:

- Implement zero-knowledge proofs for age threshold verification (e.g.,  $\text{age} \geq 18$ ,  $\text{age} \geq 21$ ).
- Ensure proofs reveal no information beyond the threshold satisfaction.
- Support multiple threshold checks without revealing exact age.

### 3. Unlinkability and Privacy:

- Implement unlinkable showing protocols preventing correlation across sessions.
- Ensure different verifications by the same user cannot be linked.
- Prevent the issuer from tracking credential usage.

### 4. Practical Security Features:

- Design a revocation mechanism for expired or compromised credentials.
- Implement anti-sharing measures to prevent credential lending.
- Ensure efficient verification suitable for mobile devices.
- Handle edge cases like leap years and time zone considerations.

## Implementation Guidelines

### Step 1: Credential Structure Design

Begin by designing your digital ID structure:

- What attributes need to be included? (birthdate, issuance date, expiry date, unique ID for revocation)
- How will attributes be committed and bound to the credential?
- What signature scheme will the issuer use?
- How will you encode dates to enable efficient range proofs?

Document your design choices and their security implications.

### Step 2: Commitment and Issuance Protocol

Implement the credential issuance system:

- Use Pedersen commitments to hide attribute values.
- Implement the issuer's signature scheme (e.g., BBS+, CL signatures, or structure-preserving signatures).
- Ensure the issuance protocol doesn't leak attribute information to the issuer beyond what's necessary.

### Step 3: Zero-Knowledge Proof Construction

Build the zero-knowledge proof system for age verification:

- Implement range proofs to prove age thresholds without revealing exact birthdate.
- Use Schnorr-style proofs or Bulletproofs for efficient range proofs.
- Ensure proof generation and verification are efficient enough for practical use.

### Step 4: Unlinkability Mechanisms

Implement privacy-preserving showing protocols:

- Design randomization techniques to make each showing unlinkable.
- Implement proof of knowledge of a valid credential without revealing which one.
- Ensure the issuer cannot track credential usage even with verifier collusion.

### Step 5: Practical Features

Add real-world functionality:

- Implement a privacy-preserving revocation mechanism (e.g., accumulators or revocation lists with zero-knowledge membership proofs).
- Add anti-sharing measures (e.g., binding to device-specific secrets or biometric templates).
- Optimize for mobile device constraints.

## Deliverables

Submit the following:

1. Source code for your privacy-preserving age verification system.
2. Design document including:
  - Detailed protocol specifications for issuance and verification.
  - Description of cryptographic primitives and their security properties.
  - Privacy analysis showing unlinkability and selective disclosure guarantees.
3. Performance analysis report with:
  - Benchmarking results for proof generation and verification times.
  - Storage requirements for credentials and revocation data.
  - Scalability analysis for different deployment scenarios.
4. Security and privacy analysis discussing:
  - Formal security properties achieved by your system.
  - Privacy guarantees and potential information leakage.
  - Comparison with existing age verification approaches.
  - Limitations and potential improvements.

## Evaluation Criteria

Your project will be evaluated based on:

- Correctness of cryptographic implementations and protocols.
- Strength of privacy guarantees and unlinkability properties.
- Practicality of the solution for real-world deployment.
- Effectiveness of revocation and anti-sharing mechanisms.
- Quality of code and documentation.

## Resources

- Steven M. Bellovin, *Privacy-Preserving Age Verification—and Its Limitations*, Columbia University, 2025.
- Alan Stapelberg, *Opening up 'Zero-Knowledge Proof' technology to promote privacy in age assurance*, Google Blog, 2025.
- Course materials on zero-knowledge proofs.

## Submission Guidelines

- Submit your code as a ZIP archive or through a Git repository.
- Include all documentation in PDF or Markdown format.
- Presentations: Prepare a 10-minute presentation demonstrating your privacy-preserving age verification system and its privacy guarantees.